Ariane Final Report

U8-F4: TESTA

At the start of our mandate, our main goal was to reduce Testa's dependence on Ski Elite and build a more balanced, resilient business. We identified two growth opportunities. The first was Ski 2000, aimed at high-volume sales in the entry-level market. The second was Raquette+, designed to keep sales and cash flow steady in spring and summer, when ski demand drops. Beyond adding revenue streams, this strategy improved how we used our production capacity. We balanced the workload between the MX and MY machines, secured volume discounts on fibre and resin, and gradually increased the share of recycled materials across our product lines.

We implemented this strategy step by step. In Period 1, we produced a small test batch of Ski 2000 to gauge how the entry-level market would respond to our pricing. Meanwhile, we kept Ski Elite production high to cover fixed costs and secure early cash flow. From Period 2 onward, we launched Raquette+ and steadily raised the proportion of recycled fibre and resin in our products. In parallel, we invested regularly in eco-friendly processes and better working conditions (QVT). These efforts reduced our environmental impact and led to our CSR certification in Period 8.

SUMMARY OF STRATEGY AND DECISIONS:
P1 - test batch of ski2000 production to explore the new market; kept the production of ski elite high to have better results. Same advertising expenditure on skiElite to not lose market share.
P2 - ski2000 market responded well, we expanded in it. We decided to reduce the production of Ski elite to establish ourselves in the ski 2000 market. We sampled the racket market as well. Given the high demand and the low offer we made ski-elite prices higher and we still managed to sell most of them
P3 - Lower price, lower production of skielite, same for ski2000 to leave space for racket production due to peak demand. We also rizing the prices of raclette to create good margins.
P4- 40% of all year sales (on the SKI market) were on P4, so we increased advertising on SKI2000 (SKI2000 market react well to increase on publicity). Higher price for ski elite and ski 2000 to match the increasing price of the Raw material to conserve good margins.
P5- Increased the prices of both ski elite and ski2000, reduced the production on ski elite to do a better repartition of the production units between the ski and rackette. Increasing our stock to match the higher demand of the P6 on ski and on rackette.
P6 - Highest demand: dump all accumulated stocks in the market, while still producing and selling skis
P7 - The firm is still lacking under the environmental POV. We pushed for investments and gave dividends out, given the high cash volume we had. The idea was to accept a net negative cashflow while spending on social and environmental indicators strengthening. The strategy worked under this point of view, allowing us to get environmentally certified in the next quarter and boost sales. We underestimated the impact of this on our average ROE, which was penalized due to the negative ROE we got in this quarted
P8 - The objective was to boost ROE and dump stocks, while maintaining a high environmental score. To do this we gave huge dividends (10M) and tried to sell just a bit shy of the expected demand of skis of both types optimizing for cost* expected sales volume. We also reduced investments to maximize profit and therefore boost economic indicators

How did we account for demand?

To ensure optimal production planning and inventory management, our firm implemented a dynamic, multiplicative demand forecasting model. Our model calculates future demand by adjusting an historical baseline against seasonal fluctuations and our strategic decisions relative to the market.

The core algorithm used to predict demand for a given period (t) is defined by the following formula:

$$Demand_t = TrueDemand_{t-1} \times \left(\frac{Seasonality_t}{Seasonality_{t-1}}\right) \times \left(\frac{Price_t}{CompetencePrice_t}\right)^{E_p} \times \left(\frac{Marketing_t}{Marketing_{t-1}}\right)^{E_m} \times \left(\frac{SalesForce_t}{SalesForce_{t-1}}\right)^{E_s}$$

Where:

- **TrueDemand:** The actual market demand from the previous period, strictly adjusted for any Out-Of-Stock (OOS) percentages to reflect unmet customer needs rather tan production constraints.
- **Seasonality:** The industry's cyclical demand coefficients for winter, spring, summer, and autumn.
- **Competence Price:** The weighted average price of all competing firms in the specific product category, isolating market shares.

- E_p, E_m, E_s **:** The calibrated elasticities for Price, Marketing, and Sales Force, respectively.

During the initial periods of the simulation, we recognized that relying solely on theoretical elasticities would lead to significant forecasting errors. Consequently, we adopted a data-driven approach, utilizing the *Ski Survey* and *Racket Survey* reports to reverse-engineer the underlying market dynamics.

By cross-referencing the performance of all five firms, isolating specific variables (e.g., comparing firms with identical sales forces and product quality but different price points or advertising budgets), we successfully calibrated our elasticities:

- **Price Elasticity (EpEp):** Initially estimated at a conservative -1.4, empirical data from Period 1 revealed that the market was highly sensitive to price deviations for identical technical coefficients. We adjusted our models to reflect a much steeper elasticity (ranging from -3.3 to -4.75 depending on the product). This insight prevented us from severely overpricing our products and losing market share to cheaper competitors.
- **Marketing Elasticity (EmEm):** By analyzing isolated marketing investments across competitors, we calibrated this parameter to approximately 0.24 - 0.38, allowing us to calculate the exact ROI of our advertising campaigns and avoid overspending in mature markets.
- **Sales Force Elasticity (EsEs):** Calibrated precisely at 0.328. Furthermore, we refined our inputs to account for *effective* salespeople, recognizing that new hires operate at a 50% efficiency rate during their first quarter of adaptation.

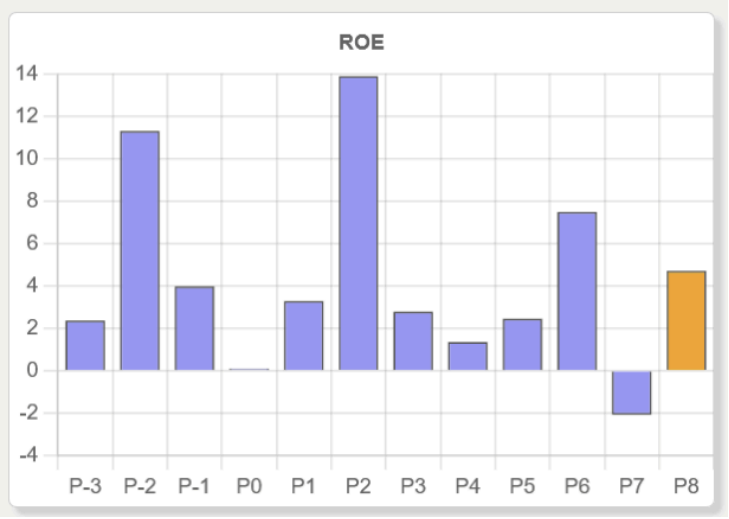
To refine our forecasts, we incorporated three critical market phenomena:

- **OOS & True Demand:** We used TrueDemand=Sales×(1+%OOS)TrueDemand=Sales×(1+%OOS) to capture the actual market size, avoiding underestimations caused by competitor production caps.
- **Spillover Effect:** We isolated our "pure" primary demand from artificial sales spikes caused by competitor stockouts, preventing subsequent over-manufacturing.
- **Launch Penalties:** We factored in the 50% first-quarter distribution penalty for new products, applying a 2x multiplier in subsequent periods to reflect full market penetration.

This data-driven framework minimized excess inventory costs and maximized value during seasonal peaks.

RESULTS ANALYSIS

ROE - Our mean ROE on the 7 first periods is 4.17 - While modest, this positive result demonstrates our ability to maintain overall profitability despite significant strategic investments.



RSE - Our mean RSE is 0.26 - We invested very late on ecological production and on recycled materials. We focused more on quality of life at work and the satisfaction of our workers by managing extra time.
Global Impact - Our final Global compact score is 0.43 - It reflects the radical and deliberate choices we had to make in order to transform the company. Our massive €10 million investment in Period 7 to secure ISO 14001 certification caused an anticipated, temporary net loss that the evaluation algorithm penalized as a 'financial irregularity'.

Sommaire Firme 4 / Période 8

Firmes	C.A. (K€)	Résultat (K€)	Trésorerie (K€)	ROE Moyen	RSE Moyen	Global Compact
Firme 1	61 148	3 808	-8 622	6,40	0,27	0,64
Firme 2	49 294	-531	-3 846	1,76	0,22	-0,69
Firme 3	40 531	734	-3 958	-1,05	0,46	-0,30
Firme 4	66 799	3 248	6 888	4,24	0,29	0,22
Firme 5	43 323	1 941	20 924	1,45	0,42	0,13

Evaluation of objectives and differences between expected results and actual results:
The Period 8 results reveal an interesting contrast between our initial forecasts and the firm's actual operational reality. On one hand, total sales revenue reached €66.79 million, falling 10.90% short of our €74.08 million forecast. This negative variance is explained by the *Raquette+* product line: while the model's estimates projected demand exceeding 70,000 units, actual sales only reached 57,660 units, leaving a surplus of 40,451 unsold rackets in inventory. Conversely, both *Ski Elite* and *Ski 2000* sold 100% of their manufactured units (100,000 and 117,000 units respectively, ending with zero inventory), which suggests we experienced stockouts and lost potential sales in these categories by not producing enough to meet the true market demand. Paradoxically, despite this drop in overall revenue, our net profit surged to €3.24 million, significantly outperforming the €2.30 million forecast. This substantial improvement in profitability was directly driven by cash outflows being €2.16 million lower than planned, combined with the cost savings derived from the strategic decision to halt new racket production during the quarter, which strongly contained our overall variable production expenses.

Next 2 years strategy:
The expansion in the ski2000 80% fiber market is clearly the way to go. The company should maximize its presence in the highest profit margin markets (skiElite and ski2000) while selling off the remaining stock of rackets. rackets should be used as a filler product: max ski production and when you can produce more than the expected demand then and only then produce rackets. accumulate them during the year to sell them around highest demand. Simultaneously grow production capacity.